

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 46

CLIMATE CHANGE - 7

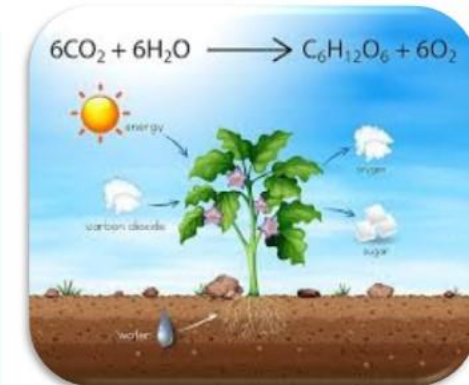
2 NATIONAL MISSION FOR A GREEN INDIA

OVERALL GOAL

The Mission aims at responding to climate change by enhancing carbon sinks.

MISSION OBJECTIVES

1. Increased forest/tree cover on 5 M ha of forest/non-forest lands and improved quality of forest cover on another 5 M ha of non-forest/forest lands.
2. Improved ecosystem services including biodiversity, water cycle and carbon sequestration from the forested lands.
3. Enhanced annual CO₂ sequestration by 50 to 60 million tonnes in the year 2022.
4. Increased forest-based livelihood income for about 3 million households, living in and around the forests.



MISSION TARGETS

1. Improvement in **quality of forest cover** and ecosystem services from forests
2. Eco-restoration and afforestation of scrub, **shifting cultivation areas**, degraded mangroves, ravines and abandoned mining areas.
3. Improvement in **forest and tree cover in urban and peri-urban lands**.
4. Improvement in forest and tree cover on marginal agricultural lands and other non-forest lands under agro forestry.

THE CORE PRINCIPLES

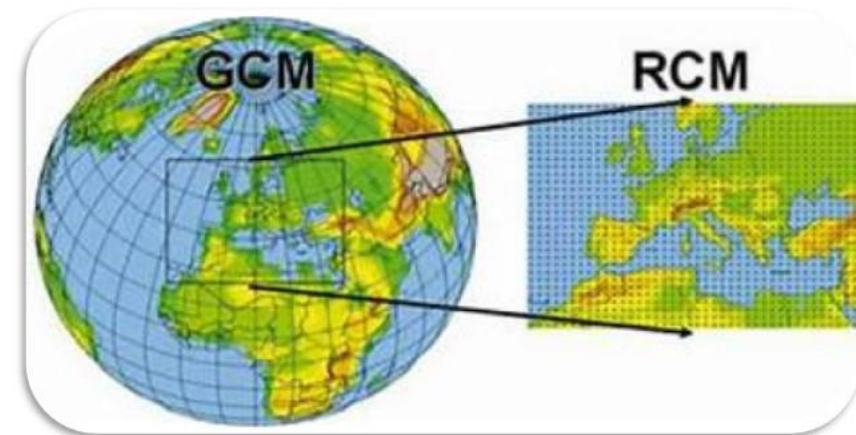
1. The first right on the forest goods and services would be that of the local communities, as mandated in the **National Forest Policy (NFP) of 1988**.
2. The Mission would contribute to empowerment of communities and **strengthen the decentralized local governance of forests**.
3. The scope of greening will not be limited to just trees and plantations; **emphasis will be on restoration of ecosystems and habitat diversity**.
4. The Mission will try to **enhance ecosystem goods and services** and carry out biodiversity conservation.

3 NATIONAL MISSION ON STRATEGIC KNOWLEDGE FOR CLIMATE CHANGE (NMSKCC)

The Mission seeks to build a dynamic knowledge system that would support national action for responding effectively to climate change.

WHAT IS REQUIRED TO RESPOND EFFECTIVELY TO CLIMATE CHANGE?

1. Region specific modeling.
2. Assessment of existing and alternate technologies.
3. International cooperation for selection and development of new technologies for adaptation and mitigation and also ensuring that knowledge gaps are bridged.



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NATIONAL MISSION ON STRATEGIC KNOWLEDGE FOR CLIMATE CHANGE (NMSKCC)

WHY IS THE MISSION NEEDED?

1. Existing knowledge gaps in many aspects of climate change.
 2. Insufficient observational and scientific information data base.
 3. Inadequate knowledge on the impacts of climate change in different areas - agriculture, forests, health, biodiversity, coastal areas etc.
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1. Poor connectivity between research institutions.
 2. Lack of an institutional mechanism for collating, synthesizing and delivering information for decision-making.
 3. Absence of a strong and vibrant global technology watch system.
 4. Absence of knowledge to make proper selection of technology for meeting the developmental goals.



3 NATIONAL MISSION ON STRATEGIC KNOWLEDGE FOR CLIMATE CHANGE (NMSKCC)

OBJECTIVES OF THE MISSION

1. Formation of knowledge networks among the research institutions engaged in research on climate science to facilitate data sharing and exchange through a suitable policy.
2. Establishment of global technology watch groups to carry out research on proper technology selection.
3. Development of national capacity for modeling the regional impact of climate change.
4. Encouraging research in the areas of climate change impacts on agriculture, health, forests, bio-diversity, coastal zones, etc.



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NATIONAL MISSION ON STRATEGIC KNOWLEDGE FOR CLIMATE CHANGE (NMSKCC)

OBJECTIVES OF THE MISSION

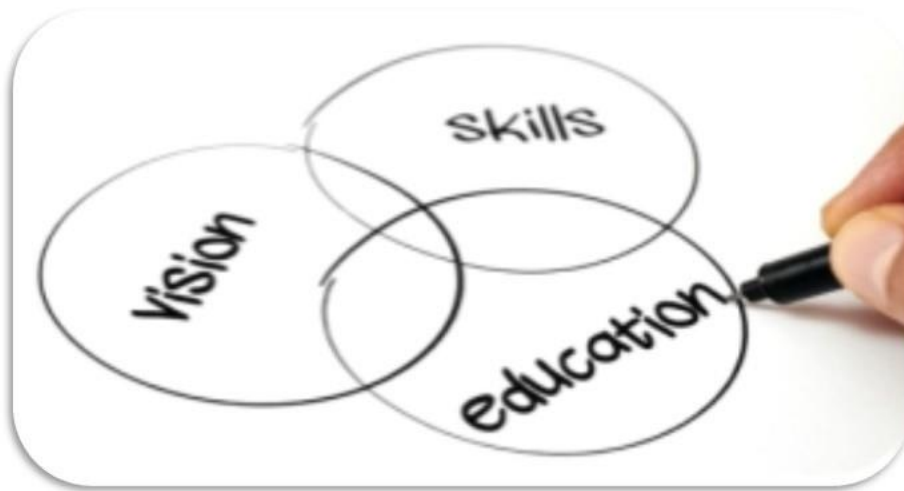


5. Generation and development of the conceptual and knowledge basis for defining sustainability of development pathways in the light of responsible climate change related actions.
6. Providing an understanding and awareness of climate risks, adaptation and mitigation practices.
7. Supporting the efforts under the other national missions and strengthening indigenous capacity for the development of appropriate technologies for responding to climate change.



3 NATIONAL MISSION ON STRATEGIC KNOWLEDGE FOR CLIMATE CHANGE (NMSKCC)

- The Mission will further address two important issues
 1. Building human and institutional capacity at various levels, and
 2. Engaging and benefiting from global collaboration in scientific research & technology development.



4 NATIONAL SOLAR MISSION

- The Jawaharlal Nehru National Solar Mission has set the ambitious target of deploying 1,00,000 MW (1GW) of solar power by 2022.
- This target is part of the overall renewable energy capacity target of 175GW, to be achieved by 2022.
- The nodal implementing agency of the Mission is Ministry of New and Renewable Energy (MNRE)

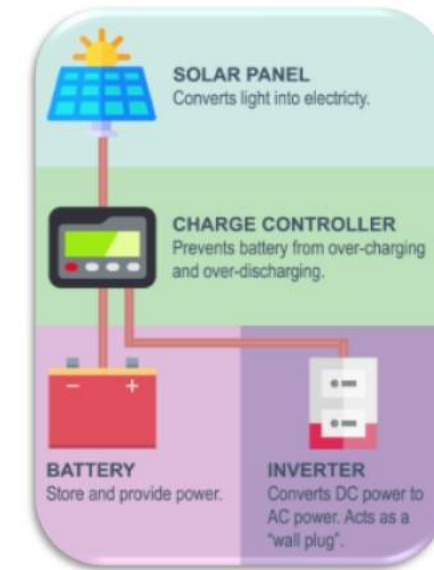


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NATIONAL SOLAR MISSION

OBJECTIVES

- The objective of the National Solar Mission is to establish India as a global leader in solar energy, by creating the policy conditions for its diffusion across the country as quickly as possible.
- The Mission seeks to create an enabling environment for solar technology penetration in the country both at a centralized and decentralized level.
- Also the Mission aims to reduce the cost of solar power generation in India through
 - a) Large scale generation
 - b) R&D and
 - c) Domestic production of raw materials and components.



4 NATIONAL SOLAR MISSION

OBJECTIVES

- The Mission also aims to deploy 20 million solar lighting systems for rural areas by 2022.
- The Mission adopts a 3 - phase approach –
 - ✓ Phase 1 (2012 - 13)
 - ✓ Phase 2 (2013 - 17) and
 - ✓ Phase 3 (2017 - 22)each phase has its own set of measurable targets.
- The target of 1,00,000 MW by 2022 will be achieved through -
 1. Decentralized Rooftop Solar Power Systems scheme (40GW)
 2. Large and Medium Grid Connected Solar Power Projects scheme (60GW) (Solar Parks Scheme)



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NATIONAL SOLAR MISSION

SOLAR PARKS SCHEME



- Initially in 2014, under NSM government planned to establish at least 25 solar parks to generate 20 GW of solar power by 2020. In 2017, the government approved the enhancement of target from 20GW to 40GW.
- So at least 50 solar parks each with capacity of 500MW or more have to be set up by 2019-2020.
- The solar parks will be developed in collaboration with the State Governments.
- Solar Energy Corporation of India (SECI) would be MNRE's agency for handling this Scheme.
- The States, applying under the scheme, will have to designate an agency (public or private) for the development of solar park (Solar Park Developer).
- The choice of Solar Park Developer for developing and maintaining the park is left to the State Government.



4 NATIONAL SOLAR MISSION

SOLAR PARKS SCHEME

- Power Purchase Agreement should be signed by the solar park developer with state and central government utilities who purchases power. Price will be set in accordance with the regulations of Central Electricity Regulation Authority.
- The Scheme proposes to provide financial support by Government of India to create the necessary infrastructure at new solar power projects.
- Incentives in the form of allocation of land, transmission and evacuation lines, access roads, availability of water and others, can be made under the project.
- Under the scheme smaller parks of 100 MW and above may be considered in NE, HP, Uttarakhand and J&K. Smaller parks of 100 MW and above may also be considered in UT's and small States with population less than 2 crores.



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